Dmytro Kuzmenko, Ph.D.

Assistant Professor at NaUKMA
Research Fellow at the University of Turin

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Dmytro Kuzmenko is a machine learning researcher in embodied AI. He works on world models and model-based RL, modular instruction-conditioned policies for multi-task robots, and human-robot interaction. His earlier work was on semi-supervised perception for exoskeletons and wearable locomotion systems. He received his PhD from NaUKMA in September 2026 and is now a Research Fellow at AIRlab, University of Turin, where he works on human-robot interaction and embodied AI projects, among them EEG brain-computer interfaces for robot control.

Research Interests

- World models and model-based RL, with policy distillation for small compute budgets
- Multi-task robot policies with modular instruction routing
- Safe human-robot interaction and HRI evaluation
- Brain-computer interfaces for control of different robot embodiments
- Sim-to-real calibration and data efficiency

Education

PhD in Computer Science *2023 - 2026*
National University of Kyiv-Mohyla Academy, Kyiv, Ukraine.
Advisor: Dr. Nadiya Shvai. Defended September 2026.
Dissertation: *Resource-Aware Embodied Intelligence Through the Optimisation of Data, Compute, and Trust.*

MSc in Applied Mathematics *2021 - 2023*
National University of Kyiv-Mohyla Academy, Kyiv, Ukraine.

BSc in Software Engineering *2016 - 2021*
National University of Kyiv-Mohyla Academy, Kyiv, Ukraine.

Academic & Research Appointments

AIRlab, Department of Computer Science, University of Turin Turin, Italy
Research Fellow Feb 2026 - present

Supervised by Prof. Cristina Gena. Visiting PhD student from February 2026, Research Fellow since September 2026. Main projects are *BCI-SMILE*, which connects EEG brain-computer interfaces to different robot embodiments, and *This/That*, on spatial reference resolution for vision-language models. Also worked on a 3D scene-mapping project for a museum tour-guide robot.

Department of Multimedia Systems, NaUKMA Kyiv, Ukraine
Assistant Professor Sep 2023 - present

Designed and teach seven graduate and undergraduate courses in machine learning, computer vision, reinforcement learning, and NLP. Started and run a student research program that scopes projects with students and mentors them through experiments, writing, and rebuttals.

Machine Intelligence Lab, University of Toronto Remote
Research Assistant Apr 2022 - Oct 2023

Supervised by Dr. Brokoslaw Laschowski. Efficient visual perception for robotic exoskeletons and wearable locomotion systems, including semi-supervised stair recognition for human-robot walking. Published at IROS 2023 and BioMedical Engineering OnLine 2024.

Publications

Peer-Reviewed Journal Articles

1. Kuzmenko, D., Shvai, N. "MoIRA: Modular Instruction Routing Architecture for Multi-Task Robotics." *Neurocomputing*, vol. 674, 132962, 2026.
doi:10.1016/j.neucom.2026.132962, arXiv:2507.01843

2. Kurbis, A. G., Kuzmenko, D., Ivanyuk-Skulskiy, B., Mihailidis, A., Laschowski, B. "StairNet: Visual Recognition of Stairs for Human-Robot Locomotion." *BioMedical Engineering OnLine*, 23(1):20, 2024. doi:10.1186/s12938-024-01216-0
3. Beimuk, V.†, Kuzmenko, D. "Energy Conservation for Autonomous Agents Using Reinforcement Learning." *NaUKMA Research Papers: Computer Science*, vol. 8, pp. 68-75, 2025. doi:10.18523/2617-3808.2025.8.68-75

Peer-Reviewed Conference Papers

1. Kuzmenko, D., Shvai, N. "When Robots Say No: The Empathic Ethical Disobedience Benchmark." In *Proc. ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Edinburgh, Scotland, pp. 168-176, March 2026. *Main track.* doi:10.1145/3757279.3785547, arXiv:2512.18474, code, project page
2. Kuzmenko, D., Shvai, N. "Knowledge Transfer in Model-Based RL Agents for Efficient Multi-Task Learning." In *Proc. 24th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, Detroit, MI, USA, pp. 2597-2599, May 2025. doi:10.5555/3709347.3743949, arXiv:2501.05329
3. Kuzmenko, D., Shvai, N. "Balancing Performance and Efficiency in Zero-shot Robotic Navigation." In *Proc. 19th International Conference on ICT in Education, Research and Industrial Applications (ICTERI 2024)*, Springer CCIS, 2025. doi:10.1007/978-3-031-81372-6_28, arXiv:2406.03015
4. Kuzmenko, D., Tsepa, O., Kurbis, A. G., Mihailidis, A., Laschowski, B. "Efficient Visual Perception of Human-Robot Walking Environments using Semi-Supervised Learning." In *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, MI, USA, October 2023. doi:10.1109/IROS55552.2023.10341654

Preprints

1. Kuzmenko, D., Shvai, N. "TD-MPC-Opt: Distilling Model-Based Multi-Task Reinforcement Learning Agents." arXiv:2507.01823, 2025 (under review). arxiv.org/abs/2507.01823, code

Work in Progress

1. *This/That: Spatial Reference Resolution for Vision-Language Models.* AIRlab, University of Turin; contributor. Tests how well VLMs and 3D pipelines resolve deictic reference ("this"/"that") from a person's egocentric viewpoint.

Workshop Papers and Posters

1. Rudas, V.†, Kuzmenko, D. "Can Vision-Language Models Assess Proxemic Risk from Egocentric Robot Images?" *Workshop on Embodied Multimodal Reasoning (EMR)* at the *European Conference on Computer Vision (ECCV)*, 2026. Peer-reviewed, non-archival; poster. arXiv:2608.12515
2. Spitkovska, V.†, Kuzmenko, D. "Trajectory Design and Budgeted Querying for Digital Twin Calibration." *Workshop on Curated Data for Efficient Learning (CDEL)* at the *European Conference on Computer Vision (ECCV)*, 2026. Peer-reviewed, non-archival; poster. arXiv:2608.08631
3. Kuzmenko, D., et al. "Semi-Supervised Learning for Efficient Visual Perception of Human-Robot Walking Environments." *Toronto Robotics Conference (TRC)*, 2024; *International Conference on Advanced Intelligent Robotics (ICAIR)*, 2023; *ICRA Workshop on Computer Vision in Robotics Research (CVWR)*, London, UK, 2023. researchgate.net, ICAIR pdf, CVWR pdf

† Supervised student.

Supervision & Student Research Program

Supervised **1 MSc thesis, 12 BSc theses, and 10 course papers** at NaUKMA, 2023 - 2026, in reinforcement learning, embodied perception, multi-agent systems, and vision-language models. Two of these projects became ECCV 2026 workshop papers, which the students presented as posters in September 2026.

Selected Supervised Research

- **O. Severhin**, *Efficient Policy Learning via Knowledge Distillation for Robotic Manipulation.* MSc thesis, 2025; co-supervised with Dr. Nadiya Shvai. Applies the distillation approach of TD-MPC-Opt to compute-efficient

manipulation policies.

- **V. Spitkovska**, *Active Reinforcement Learning for Data-Efficient Digital Twin Calibration.* BSc thesis, 2026. Active trajectory selection under a fixed query budget for sim-to-real parameter identification. Led to the CDEL workshop paper at ECCV 2026.
- **V. Rudas**, *Vision-Language Models for Proxemics in Robot Perception.* BSc thesis, 2026. Tests whether VLMs can tell when a person is too close, socially or physically, from the robot's own camera view. Led to the EMR workshop paper at ECCV 2026.
- **T. Matiichyk**, *Ego-Centric Vision for Hand Occupancy Detection: Toward Safer Human-Robot Interaction.* BSc thesis, 2026. Detects from an egocentric camera whether a person's hands are free, a precondition for safe handover.
- **Y. Vinokur**, *Cooperation of Partially Observed Agents in Ad-Hoc Open Teams.* BSc thesis, 2025. Benchmarked eight decentralised multi-agent RL methods under partial observability and changing team membership; ported the Wildfire benchmark to PettingZoo.
- **V. Beimuk**, *Energy Conservation in Autonomous Agents Using Reinforcement Learning.* BSc thesis, 2025; led to a co-authored journal paper. Soft Actor-Critic with a fuel penalty in the reward, to study the speed/energy trade-off in learned driving policies.
- **V. Bezborodov**, *Object Detection Model Pruning with Application to Human Recognition in UAV Footage.* BSc thesis, 2025. Quantisation, structured pruning, and distillation for real-time on-board inference.

Skills

- Robots: Reachy Mini, Booster K1, Pepper, Sanbot. Set up Reachy Mini as a shared experimental platform at AIRlab.
- Open-source research software: the EED-Gym benchmark and the TD-MPC-Opt distillation framework, both with code to reproduce the papers.
- Research methods: model-based RL, world models, policy distillation and quantisation, imitation learning, mixture-of-experts policies, VLA and VLM systems, embodied-AI benchmarking, HRI evaluation.
- Simulation: MuJoCo, Gymnasium, Isaac Gym and Isaac Lab, Genesis, PyBullet, Habitat, PettingZoo.
- Programming and ML: Python, PyTorch, Hugging Face, TensorFlow, Docker, Git, distributed training.
- Deployment: vLLM, low-latency inference, agentic pipelines, cloud deployment on Azure, AWS, and GCP.

Invited Talks & Conference Presentations

When Robots Say No: The Empathic Ethical Disobedience Benchmark Apr 2026
Presentation at the *COMETE Workshop*, University of Turin, Turin, Italy

When Robots Say No: The Empathic Ethical Disobedience Benchmark Mar 2026
Main-track paper presentation, *HRI 2026*, Edinburgh, Scotland

Knowledge Transfer in Model-Based RL Agents Jun 2025
Invited talk, RL Reading Club, *University of Southampton*, Southampton, UK

Knowledge Transfer in Model-Based RL Agents for Efficient Multi-Task Learning May 2025
Poster, *AAMAS 2025*, Detroit, MI, USA

Potential of Vision-Language-Action (VLA) Models Mar 2025
Talk, *Kyiv AI & BigData Day*, Kyiv, Ukraine

SSL for Efficient Perception of Human-Robot Walking Environments 2023 - 2024
Posters at *TRC 2024*, *ICAIR 2023*, and the *CVWR Workshop at ICRA 2023*

Industry and Public Talks
Over 20 invited talks at industry AI/ML conferences and professional meetups in Ukraine and abroad. Recordings.

Academic Service

- Conference reviewer: *HRI* 2026 and 2027, *ICSR* 2026, *ICTERI* 2026
- Workshop reviewer: *PALM* at NeurIPS 2026, *CDEL* at ECCV 2026
- Journal reviewer: *Journal of Field Robotics*, *Behaviour and Information Technology*

Teaching Experience

All courses at NaUKMA.

Course	Level	Terms	Offerings
Computer Vision	MSc	Fall 2023 - Fall 2026	4
Machine Learning	MSc	Fall 2023 - Fall 2026	4
Reinforcement Learning	MSc	Spring 2024 - Spring 2025	2
Image Analysis and Computer Vision	MSc	Spring 2024 - Spring 2026	3
NLP Systems	MSc	Fall 2024 - Fall 2026	3
Machine Learning	BSc	Fall 2024 - Fall 2026	3
Deep Learning for Computer Vision	BSc	Spring 2024 - Spring 2026	3

Industry & Engineering Experience

Over nine years of software and applied-AI engineering alongside research. Recent positions are listed below.

Head of AI, OnlyMonster *Jan 2025 - present*

First AI hire; built the AI team and chose its technical stack. Production LLM and agent systems for real-time chat assistance, multilingual translation, and content moderation, with low-latency inference.

Data Science Engineering Manager, Litslink *Nov 2022 - Dec 2024*

Led applied ML projects in computer vision, predictive modelling, and LLM pipelines, more than 25 production and proof-of-concept projects in total, including Sportsbox AI.

Machine Learning Engineer, Infopulse *Nov 2021 - Nov 2022*

Automated segmentation and OCR for presentation slides. Built a technical-diagram processing pipeline (mmdet, OpenCV) that reconstructs the diagram graph with BFS, and made its inference about 70% faster.

Languages

Ukrainian (native), English (C2), German (B2), Italian (B1).

References

Available upon request.